

RITIK SHARMA

AI / ML Engineer — Computer Vision · Generative AI · Edge Deployment

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PROFESSIONAL SUMMARY

AI/ML Engineer with nearly 2 years of experience across Machine Learning, Computer Vision, Generative AI (RAG/LLMs & agentic AI), and Edge Deployment. Architected production-grade defect-detection systems at 95%+ accuracy for global markets (US/Japan) and optimized high-speed vision pipelines with TensorRT/CUDA to cut inference latency by 98%. Takes models from data preparation and training through to real-time, edge-optimized deployment.

TECHNICAL SKILLS

Languages: Python, SQL (SQLite), Bash

Machine Learning: scikit-learn, XGBoost / LightGBM, Random Forest, SVM, regression & clustering (K-Means, PCA), feature engineering, model evaluation (cross-validation, ROC-AUC, F1), hyperparameter tuning (Optuna, GridSearch)

Deep Learning: PyTorch, TensorFlow, CNNs, RNN / LSTM / GRU, Transformers (self-attention, BERT), Autoencoders, GANs, transfer learning & fine-tuning

Computer Vision: YOLOv8 / YOLOv11s, ConvNeXt, Vision Transformers (ViT), OpenCV, object detection, defect localization, sub-pixel analysis, synthetic data generation

Generative AI / NLP: RAG pipelines, LangChain, agentic AI (agents & chatbots), Hugging Face Transformers, spaCy / NLTK, LLMs (DeepSeek R1, Mistral, Llama), embeddings, Vector DBs (ChromaDB, Qdrant)

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, EDA, data preprocessing, Jupyter

MLOps & Deployment: Docker, FastAPI, MLflow, ONNX, TensorRT, CUDA / PyCUDA, NVIDIA Jetson (edge), Kafka, REST APIs, Git, Linux, model quantization

Tools & Other: PySide6 / PyQt, Streamlit, Socket.io, Hadoop, PySpark

PROFESSIONAL EXPERIENCE

GenAI Engineer II — **HCLTech** *Pune, India* | Aug 2026 – Present

- **Focus:** Building enterprise GenAI solutions — LLM applications, RAG, and agentic AI systems — delivered on Azure cloud.

Software Developer — **Griffyn Robotech Pvt. Ltd.** *Pune, India* | Nov 2024 – Aug 2026

Project: Impersonator — Mobile Screen Defect Detection System

- **Production Deployment:** Engineered a vision system deployed across the US and Japan for mobile-screen defect detection (dead pixels, discoloration, line defects), sustaining 95%+ accuracy over thousands of production cycles.
- **Line-Defect Analysis:** Built a line-defect detection module that localizes each defect and reports its count, size/length, and orientation (horizontal vs. vertical) per screen for precise QC grading.
- **Dead-Pixel Analysis:** Developed dead-pixel detection returning per-defect coordinates, size, severity level, and detailed classification metadata for downstream reporting.
- **Model Architecture:** Migrated from legacy image processing to a deep-learning pipeline using ConvNeXt trained on 20,000+ images, and customized YOLOv11s (P2 layer) for microscopic sub-pixel defect localization.
- **Data Engineering:** Solved data scarcity with a synthetic data-generation engine that programmatically creates complex defect patterns (e.g., screen burn), saving 3+ months of manual collection.
- **Distributed Inference:** Co-architected a FastAPI inference service letting edge machines offload heavy model execution to a centralized server for real-time processing.

Project: Optivity — High-Speed Vision & GenAI

- **Hybrid Detection Pipeline:** Built an end-to-end Barcode/QR recognition pipeline on NVIDIA Jetson using YOLO localization, custom OpenCV preprocessing, and PyZbar decoding.
- **Latency Optimization:** Re-engineered detection with CUDA and TensorRT on NVIDIA Jetson (tuned for power/thermal limits), cutting inference time from 2000 ms to 40 ms (98% faster) for high-throughput lines.
- **Optivity Agent (Agentic AI):** Built an internal conversational AI assistant (chatbot + agent) using DeepSeek R1, RAG, and vector DBs over the full codebase — delivering context-aware debugging, architecture insights, and automated task assistance to the dev team.

Project: Bulk & Pouch Packaging

- **Full-Stack Development:** Developed a modular PySide6 desktop UI integrated with a high-speed Socket.io backend for real-time production data in reverse-logistics packaging systems.

Intern — **C-DAC** *Noida, India* | Sept 2022 – Feb 2023

- **Big Data Analytics:** Worked with Hadoop and PySpark to build ML prototypes for large-scale data processing.
- **Sentiment Stream:** Built an end-to-end NLP pipeline for product-review analysis using Python, LSTM, and Streamlit.

EDUCATION

PG-DBDA (PG-Diploma in Big Data Analytics & Machine Learning) — **C-DAC, Pune** 2024

B.Tech, Computer Science Engineering — **AKTU, Uttar Pradesh** 2023

INTERESTS Lo-Fi music production • Reading about astronomy